

Mathematical Induction for Natural Numbers

Lecture Notes

Course: Computer-Assisted Theorem Proving (7CCSACTP)

Context & Induction Intuition

→ **Previous Videos:**

- Basic elements of interface (terms, theorems)
- Basic proof methods (FW, BW, subst)

→ **This Video:**

- Mathematical induction for natural numbers (`nat`)

Intuition for Mathematical Induction:

- To prove a property P holds for all $n \in \mathbb{N}$ (e.g. $?x \leq ?x + 3$), we prove two cases:
 - **goal₁ (Base Case):** $P(0)$ holds.
 - **goal₂ (Step Case):** Assuming $P(n)$ holds (Induction Hypothesis, I.H.), $P(n + 1)$ holds.

- Algorithmic view of induction:

$$P(0) \xrightarrow{\text{goal}_1} P(1) \xrightarrow{\text{goal}_2} P(2) \xrightarrow{\text{goal}_2} \dots \xrightarrow{\text{goal}_2} P(n)$$

- **Isabelle Note:** In Isabelle, $n + 1$ is represented as `Suc n` for natural numbers.

Detailed Example: Commutativity of Addition ($?x + ?y = ?y + ?x$)

Proof Strategy: Induction on y .

Base Case ($y = 0$):

- **goal₁ :: $x + 0 = 0 + x$**
- **Lemmas:** `lem1:  $x + 0 = x$ ,    lem2:  $0 + x = x$`
- **Proof steps:**
 1. $x + 0 = x$ (by subst `lem1`)
 2. $\dots = 0 + x$ (by subst `lem2`)
- finally $x + 0 = 0 + x$ ✓

Step Case ($y \rightarrow \text{Suc } y$):

- **Assume I.H.:** $x + y = y + x$
- **goal₂ :: $x + (\text{Suc } y) = (\text{Suc } y) + x$**
- **Lemmas:**
 - `lem3:  $?a + (\text{Suc } ?b) = \text{Suc } (?a + ?b)$`
 - `lem4:  $(\text{Suc } ?a) + ?b = \text{Suc } (?a + ?b)$`
- **Proof steps:**
 1. $x + (\text{Suc } y) = \text{Suc } (x + y)$ (by subst `lem3`)
 2. $\dots = \text{Suc } (y + x)$ (by subst I.H.)
 3. $\dots = (\text{Suc } y) + x$ (by subst `lem4`)
- finally $x + (\text{Suc } y) = (\text{Suc } y) + x$ ✓

Isabelle Syntax Reminder

- In Isabelle: `Suc x` $\approx x + 1$ for terms of type `nat`.
- Induction rule applied via: `apply (induct y)` or `proof (induct y)`.